

ON THE IMPACT

We work together with the breeding industry to help identify innovative solutions for improving crops. Having these collaborations means that we often need to protect our findings as intellectual property.

This protection is needed because regular scientific publishing makes the findings and solutions less attractive for industry. It is important to work with companies to make sure that our knowledge eventually benefits the growers.



Lettuce field trial in the public-private partnership LettuceKnow.

ON PREVIOUS EXPERIENCES

We are a part of the CropXR Consortium, which has a Communications Office. They do an excellent job, and we sometimes provide materials to them. I also think teaching is a great way to communicate science.

In terms of science communication, I think we’re doing quite well. There are many activities where science is shared with broad audience: open days at universities, exhibitions (like this one), videos with research insights, and public events.

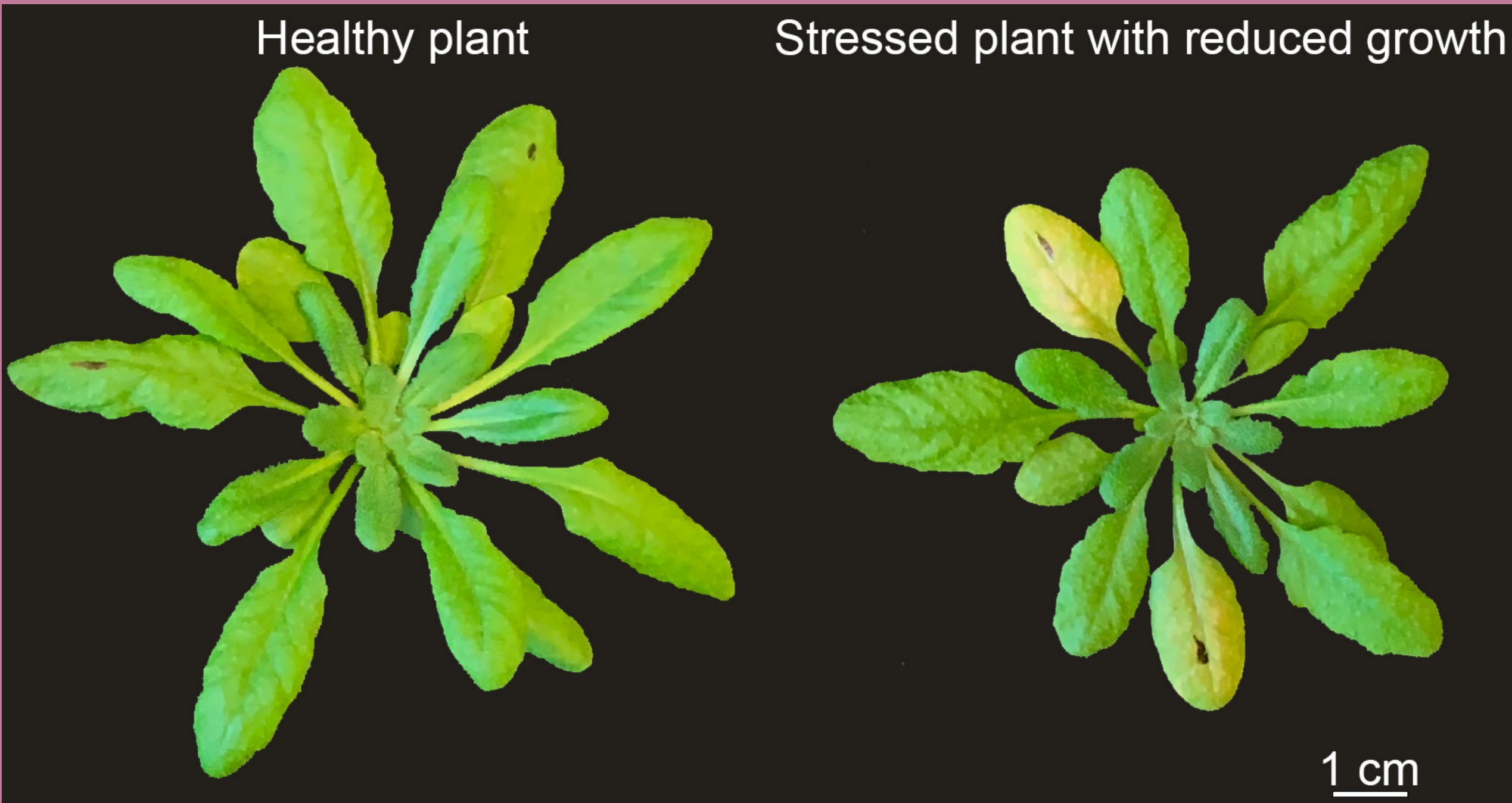
ON THE FUTURE

I think there are a few things we can do as scientists for science communication:

Explore creative outreach formats, such as art exhibitions using beautiful microscopy images, especially for more specialized or niche research areas.

Focus on storytelling by highlighting scientists as the main heroes of a story to make content more engaging and relatable.

Create more content-rich books about plants for both adults and children.



Using model plant *Arabidopsis thaliana*, we study how plants regulate their growth during and after an immune response. On the picture, a plant with an activated immune response has a smaller rosette (image by Nienke van der Wal).



Dr. Dmitry Lapin

My lab studies how crop plants like lettuce (*Lactuca sativa*) and our model plant *Arabidopsis thaliana* can grow and develop despite infections and diseases.

How do research and industry work together?



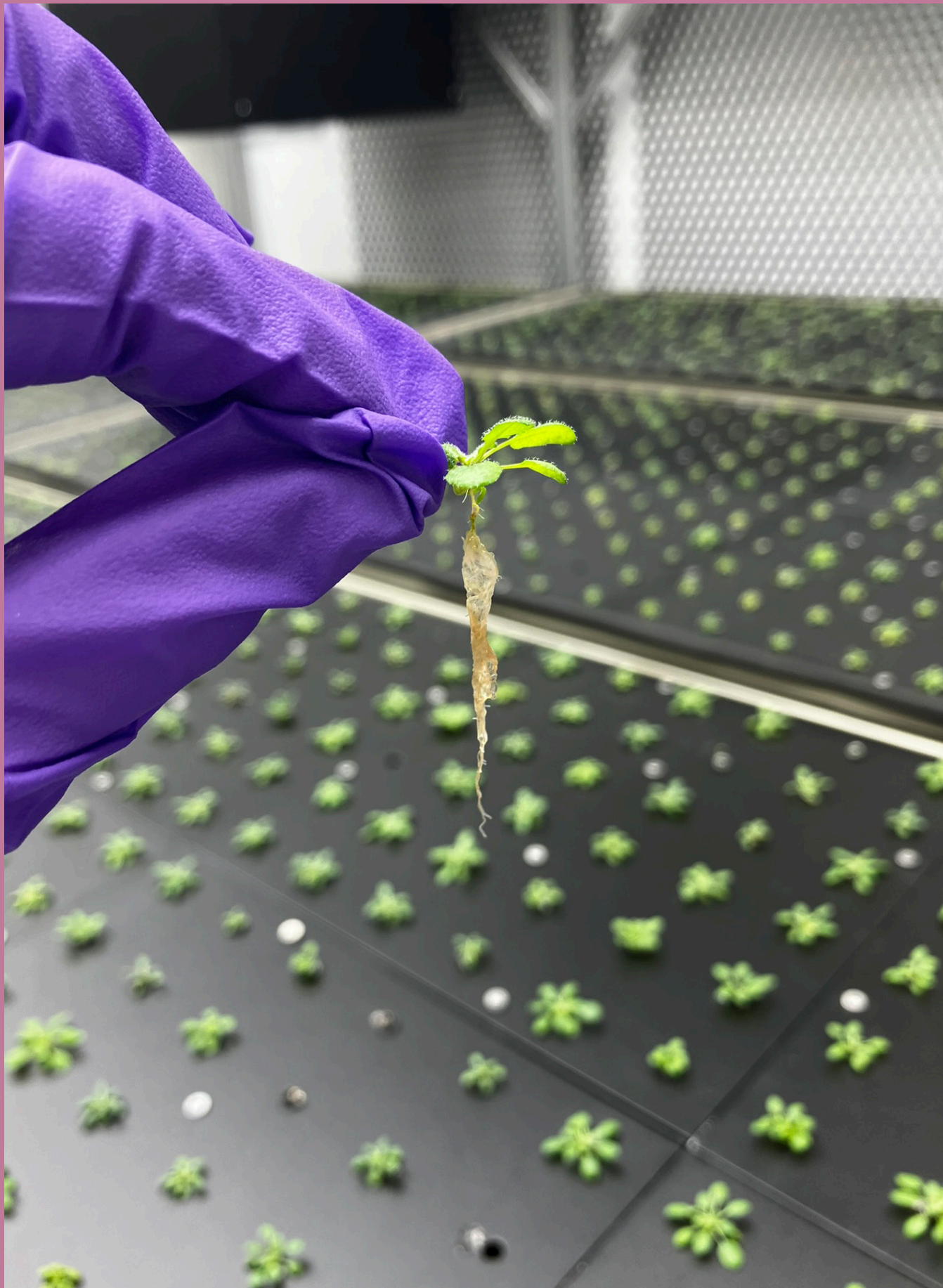
Crops such as lettuce have many diseases. Research in model plants such as *Arabidopsis thaliana* can help us find solutions against them. We use knowledge about mechanisms of *Arabidopsis* immunity to inform solutions that can help breed new varieties requiring less pesticides for plant protection. (Downy mildew in *Arabidopsis* (top) and in lettuce grown in field (bottom left) or in our laboratory (bottom right)).

ON THE CHALLENGES

I believe the general public does not need to understand every research detail to benefit from it. Just like with medical treatments, people can use technologies without knowing exactly how they work. At the same time, we should not oversimplify and do our best to explain the principles, advantages and limitations. By making everything too basic, we risk losing those who are genuinely interested in deeper understanding of plant science.



To enable translation of knowledge on immunity and growth across different species, we use tissue culture and gene editing.



High-throughput imaging of plants allows to collect a lot of information about plant growth and physiological status in an automated manner. Our laboratory works with NPEC (Netherlands Plant Ecophenotyping Center) to help understand principles of plant life and their interaction with the environment using this type of data.

Read this poster in Dutch



What do you think?

